Manuel Madeira

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About Me

Final-year PhD candidate at EPFL, advised by Pascal Frossard and Dorina Thanou. My research focuses on generative modeling, particularly controllability and the alignment of training and inference in diffusion and flow matching for discrete data. Using graph generation and language modeling as testbeds, I study both how generative models leverage the inherent structure of data and how structural and domain knowledge can be explicitly incorporated into generative processes. I apply these ideas to scientific problems, including molecular generation and digital pathology.

Education

2022 - Present — PhD in Machine Learning (Expected Graduation: December 2026)
École Polytechnique Fédérale de Lausanne, Switzerland
Research on structurally controlled graph generation, discrete diffusion and flow matching, and training–inference alignment, with applications including molecular generation and digital pathology.

2018 - 2021 — MSc. in Biomedical Engineering
Instituto Superior Técnico, Portugal
GPA: 19 / 20 (**1st** in class)
Thesis: *Using Co-Coercivity for Variance Reduction in Stochastic Convex Optimization* (**20/20**)
Supervisors: Pedro Aguiar, João Xavier, and Renato Negrinho

2018 - 2019 (Fall) — Exchange student in Computer Science
Tsinghua University, China

2015 - 2018 — BSc. in Biomedical Engineering
Instituto Superior Técnico, Portugal
GPA: 19 / 20 (**1st** in class; first student in the program to graduate with this GPA)

Experience

Apr 2026 - Sep 2026 — **Research Intern** at Apple MLR
Investigated the training–inference mismatch in masked diffusion models. Developed a trajectory-aware training approach that matched same-size autoregressive performance in small-scale experiments. Extended the approach to the 8B-parameter scale, where it enabled over 20% faster inference.

Jul 2025 - Dec 2025 — **Applied Scientist Intern** at Amazon
Developed a graph-based deep learning framework achieving an approximately 1000× efficiency improvement over existing tools for supply chain long-term planning.

Sep 2021 - Aug 2022 — **Machine Learning Researcher** at Inductiva (Startup)
Conducted research on deep-learning-based approaches to solve partial differential equations. Deployed physics-informed neural networks to model heat diffusion and coastal dynamics and analyzed their generalization to arbitrary domains.

Mar 2021 - Jun 2021 — **Research Intern** at Institute for Systems and Robotics
Extended my MSc thesis work on exploiting L-smoothness in first-order stochastic optimization, contributing theoretical developments to algorithmic analysis.

Sep 2019 - Feb 2020 — **Research Intern** at IST-ID
Transcribed interviews on health technology assessment in Portugal for the Medi-value Project.

Awards and Honors

2025 **NeurIPS 2025 Top Reviewer**
2025 **Qualcomm Innovation Fellowship Finalist**
2022-23 **EDIC PhD Fellowship**, EPFL
2016-18 **1st-ranked student across all years of the Biomedical Engineering BSc**, Instituto Superior Técnico
2015-20 **Diploma of Academic Excellence (Top 10%)**, Instituto Superior Técnico
2009-15 **Best Student in High School Award (monetary prize)**, Crédito Agrícola

Publications

* Equal contribution

[1] **MM**, A. Shidani, A. Bizeul, V. Turrisi, L. Béthune, B. Devnani, D. Busbridge, P. Ablin, and J. Monteiro. **On Trajectory-Aware Training for Masked Diffusion Language Models**. Under review. [PDF]

[2] A. Deschenaux, N. Ilieva, J. Deschenaux, **MM**, D. Thanou, A. Lucchi, A. Orvieto, C. Gulcehre, and P. Frossard. **Loopholing All the Way: Continuous Looping with Discrete Diffusion Models**. Oral Presentation. *BeNTo Workshop*. NeurIPS 2026. [PDF]

[3] S. Rozada, Y. Qin, **MM**, P. Frossard, and A. Ribeiro. **DiPhon: Diffusion on Graphons for Scalable Graph Generation**. NeurIPS 2026. [PDF]

[4] S. Öğüt, C. Vincent-Cuaz, **MM**, et al. **Benchmarking Instance-Level Learnability and Interpretability in Multiple Instance Learning**. Under review.

[5] A. Carballo-Castro, **MM**, Y. Qin, D. Thanou, and P. Frossard. **Generating Directed Graphs with Dual Attention and Asymmetric Encoding**. ICLR 2026. [PDF]

[6] B. Chen*, C. Vincent-Cuaz*, L. A. Schoenpflug*, **MM***, et al. **Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology**. Early Acceptance (Top 9%). MICCAI 2025. [PDF]

[7] O. Zaghen, **MM**, L. Toni, and P. Frossard. **Graph Discrete Diffusion: a Spectral Study**. *DeLTa and XAI4Science Workshops*. ICLR 2025. [PDF]

[8] L. Sbicego, S. Öğüt, **MM**, Y. Qin, D. Thanou, and P. Frossard. **On the Role of Structure in Hierarchical Graph Neural Networks**. *ICBINB Workshop*. Selected for PMLR Proceedings. ICLR 2025. [PDF]

[9] Y. Qin*, **MM***, D. Thanou, and P. Frossard. **DeFoG: Discrete Flow Matching for Graph Generation**. Oral Presentation (Top 1%). ICML 2025. [PDF]

[10] **MM**, C. Vignac, D. Thanou, and P. Frossard. **Generative Modelling of Structurally Constrained Graphs**. NeurIPS 2024. [PDF]

[11] **MM**, D. Thanou, and P. Frossard. **Tertiary Lymphoid Structures Generation through Graph-based Diffusion**. *GRAIL Workshop*. MICCAI 2023. [PDF]

[12] S. Moalla*, **MM***, L. Riccio*, and J. Lee*. **[Re] Reproducibility Study of Behavior Transformers**. *ML Reproducibility Challenge 2022*. Outstanding Paper Award (Honorable Mention). ReScience C 2023. [PDF]

[13] **MM**, R. Negrinho, J. Xavier, and P. M. Aguiar. **COCO Denoiser: Using Co-Coercivity for Variance Reduction in Stochastic Convex Optimization**. *OPT Workshop*. NeurIPS 2021. [PDF]

Talks

ICML 2025, Vancouver, Canada. *DeFoG: Discrete Flow Matching for Graph Generation.*

AI Alliance: AI in Materials & Chemistry Webinar Series, online, 2025. *DeFoG: Discrete Flow Matching for Graph Generation.*

Inductiva Machine Learning for Science & Engineering Summer School, Porto, Portugal, 2025. *An Introduction to Graph Neural Networks.*

BioML Workshop, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science.*

Applied Machine Learning Days, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science.*

DataMakers Fest, Porto, Portugal, 2023. *Generative Models in Science: A Peek into Graph Diffusion Models for Digital Pathology.*

Teaching Experience

Teaching:	Network Machine Learning (three editions; Lead TA in 2024/25 and 2025/26)
	Probability and Statistics
	Practice of Object-Oriented Programming
	Analysis I
	Histology
Supervision:	Six students, including one junior PhD student

Skills

Programming:	Python, Java, MATLAB, Mathematica, R
ML & Data:	PyTorch, PyTorch Geometric, Lightning, TensorFlow/Keras, NumPy, scikit-learn, Pandas
Opt. & Research:	Hydra, Weights & Biases, Git, CVXPY, CVXOPT
Infrastructure:	AWS (EC2, S3), Docker, Kubernetes, Slurm, Run:AI
Languages:	Portuguese (native), English (C2; TOEFL iBT: 114 / 120), Spanish (A2), French (A2), Chinese (elementary; Tsinghua course)

Community Service

Reviewing:	Journals: JMLR; IEEE TSIPN Conferences: NeurIPS (2025); ICLR (2026) Workshops: NPGML (NeurIPS 2025); ComBayNS (IJCNN 2025)
Outreach:	Through Native Scientists, introduced AI, generative modeling, and diffusion models for Portuguese-speaking students from migrant backgrounds in Switzerland (ages 6–18)

Extracurricular Activities

2023	Attended Cambridge Ellis Unit Summer School on Probabilistic Machine Learning
2022 - 2024	Co-organizer of Deep Learning Sessions Portugal
2021	Attended Lisbon Machine Learning Summer School
2009-21	Competitive football player (national level)
2015	National finalist in Biology Olympiad

Personal Interests

Football, CrossFit, Skiing, Reading, Cinema